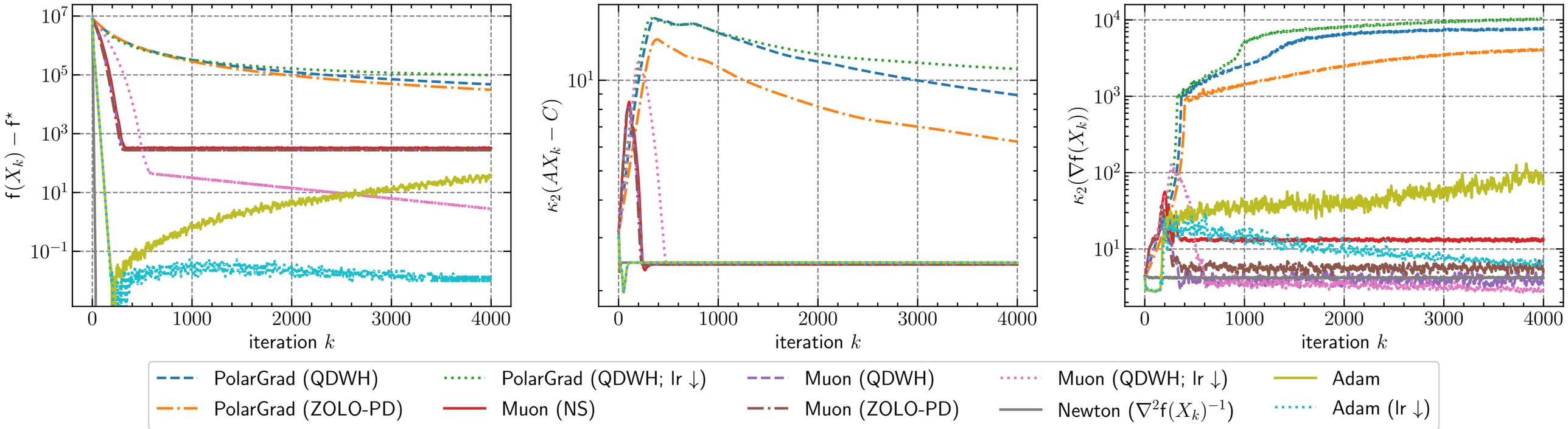
10^7
10^5
10^3
10^1
10^{-1}
$\mathsf{f}(X_k) - \mathsf{f}^\star$
0
1000
2000
3000
4000
iteration k
10^1
$\kappa_2(AX_k - C)$
0
1000
2000
3000
4000
iteration k
10^4
10^3
10^2
10^1
$\kappa_2(\nabla \mathsf{f}(X_k))$
0
1000
2000
3000
4000
iteration k
PolarGrad (QDWH)
PolarGrad (ZOLO-PD)
PolarGrad (QDWH; lr ↓)
Muon (NS)
Muon (QDWH)
Muon (ZOLO-PD)
Muon (QDWH; lr ↓)
Newton ($\nabla^2 \mathsf{f}(X_k)^{-1}$)
Adam
Adam (lr ↓)